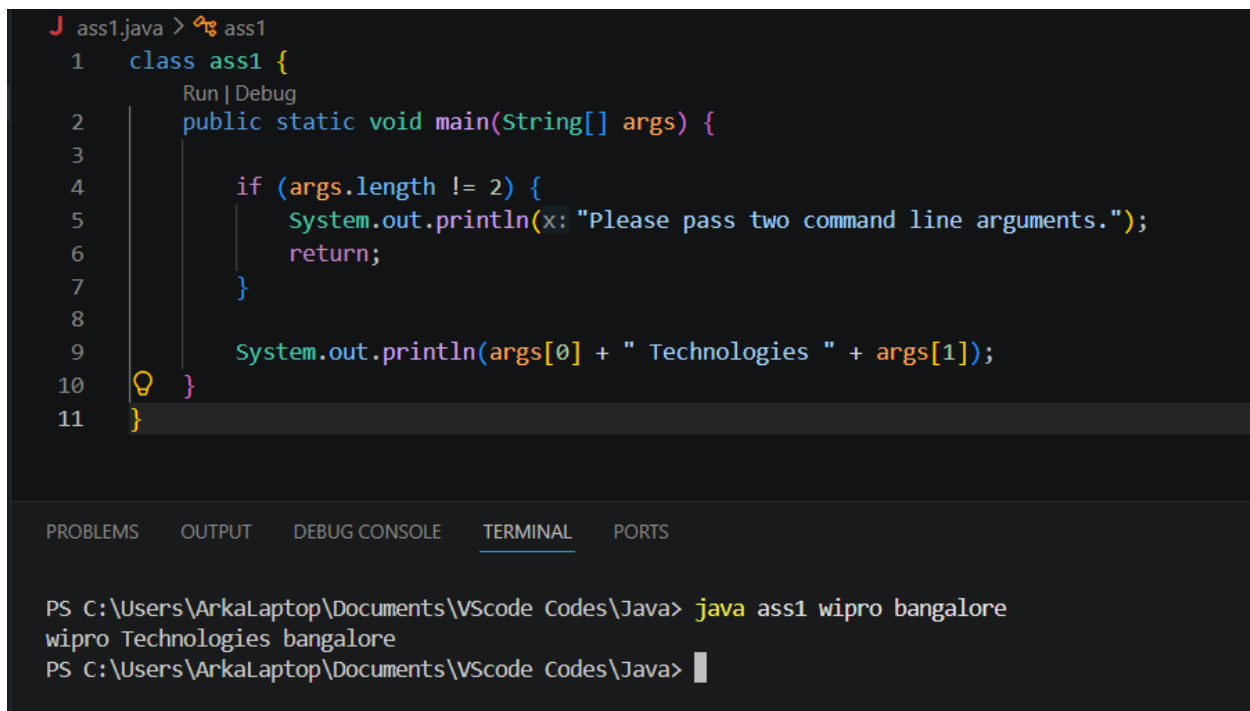


Java Fundamentals

Language Basics

Q1. Write a Java program that accepts **two Strings as command-line arguments** and generates the output in the required format.

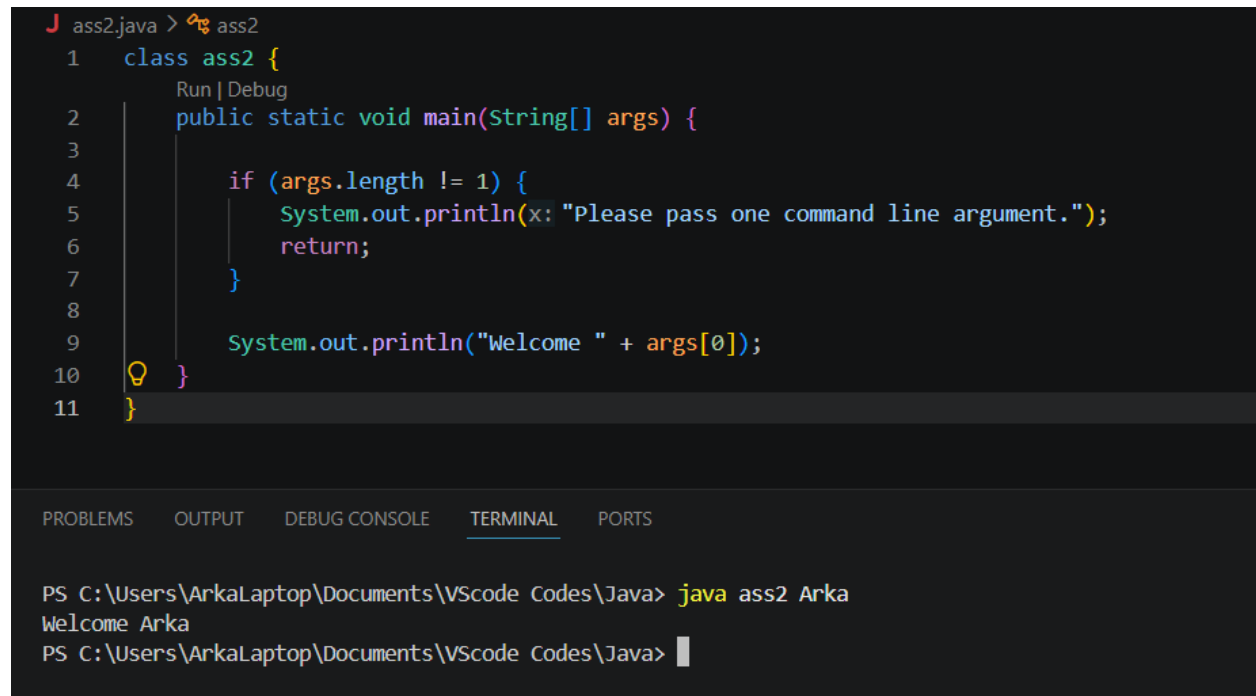


```
J ass1.java >  ass1
1  class ass1 {
2      Run | Debug
3      public static void main(String[] args) {
4          if (args.length != 2) {
5              System.out.println(x: "Please pass two command line arguments.");
6              return;
7          }
8
9          System.out.println(args[0] + " Technologies " + args[1]);
10     }
11 }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\ArkaLaptop\Documents\VScode Codes\Java> java ass1 wipro bangalore
wipro Technologies bangalore
PS C:\Users\ArkaLaptop\Documents\VScode Codes\Java> 
```

Q2. Write a Java program to accept a **String** as a **command-line argument** and print a **Welcome** message as given below.



```
J ass2.java >  ass2
1  class ass2 {
    Run | Debug
2      public static void main(String[] args) {
3
4          if (args.length != 1) {
5              System.out.println(x: "Please pass one command line argument.");
6              return;
7          }
8
9          System.out.println("Welcome " + args[0]);
10     }
11 }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\ArkaLaptop\Documents\VScode Codes\Java> java ass2 Arka
Welcome Arka
PS C:\Users\ArkaLaptop\Documents\VScode Codes\Java> 
```

Q3. Write a Java program to accept **two integers as command-line arguments** and print the **sum of the two numbers**.

```
ass3.java > ass3
1  class ass3 {
    Run | Debug
2      public static void main(String[] args) {
3
4          if (args.length != 2) {
5              System.out.println("Please pass two integer command line arguments.");
6              return;
7          }
8
9          int num1 = Integer.parseInt(args[0]);
10         int num2 = Integer.parseInt(args[1]);
11
12         int sum = num1 + num2;
13
14         System.out.println("The sum of " + num1 + " and " + num2 + " is " + sum);
15     }
16 }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\ArkaLaptop\Documents\VScode Codes\Java> javac ass3.java
PS C:\Users\ArkaLaptop\Documents\VScode Codes\Java> java ass3 10 20
The sum of 10 and 20 is 30
PS C:\Users\ArkaLaptop\Documents\VScode Codes\Java> 
```

Flow Control Statements:

Q4. A) Write a Java program to check whether a given integer is **Positive**, **Negative**, or **Zero**.

B) Given two non-negative integer values, print true if they have the same last digit, otherwise print false.

```
J ass4.java >  ass4
1  import java.util.Scanner;
2
3  class ass4 {
4      Run | Debug
5      public static void main(String[] args) {
6
7          Scanner sc = new Scanner(System.in);
8
9          System.out.print(s: "Enter an integer: ");
10         int num = sc.nextInt();
11
12         if (num > 0) {
13             System.out.println(x: "Positive");
14         } else if (num < 0) {
15             System.out.println(x: "Negative");
16         } else {
17             System.out.println(x: "Zero");
18         }
19
20         sc.close();
21     }
}
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\ArkaLaptop\Documents\VScode Codes\Java> javac ass4.java
PS C:\Users\ArkaLaptop\Documents\VScode Codes\Java> java ass4 25
Enter an integer: 25
Positive
```

```
J ass5.java >  ass5
1  import java.util.Scanner;
2
3  class ass5 {
    Run | Debug
4      public static void main(String[] args) {
5
6          Scanner sc = new Scanner(System.in);
7
8          System.out.print(s: "Enter first number: ");
9          int num1 = sc.nextInt();
10
11         System.out.print(s: "Enter second number: ");
12         int num2 = sc.nextInt();
13
14         if ((num1 % 10) == (num2 % 10)) {
15             System.out.println(x: "true");
16         } else {
17             System.out.println(x: "false");
18         }
19
20         sc.close();
21     }
22 }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\ArkaLaptop\Documents\VScode Codes\Java> javac ass5.java
PS C:\Users\ArkaLaptop\Documents\VScode Codes\Java> java ass5
Enter first number: 18
Enter second number: 23
false
```

Q5. Write a Java program to check whether a given integer number is **Odd** or **Even**.

```
J ass6.java >  ass6
1  import java.util.Scanner;
2
3  class ass6 {
4      Run | Debug
      public static void main(String[] args) {
5
6          Scanner sc = new Scanner(System.in);
7
8          System.out.print(s: "Enter an integer: ");
9          int num = sc.nextInt();
10
11         if (num % 2 == 0) {
12             System.out.println(x: "Even");
13         } else {
14             System.out.println(x: "Odd");
15         }
16
17         sc.close();
18     }
19 }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\ArkaLaptop\Documents\VScode Codes\Java> javac ass6.java
PS C:\Users\ArkaLaptop\Documents\VScode Codes\Java> java ass6
Enter an integer: 456
Even
```

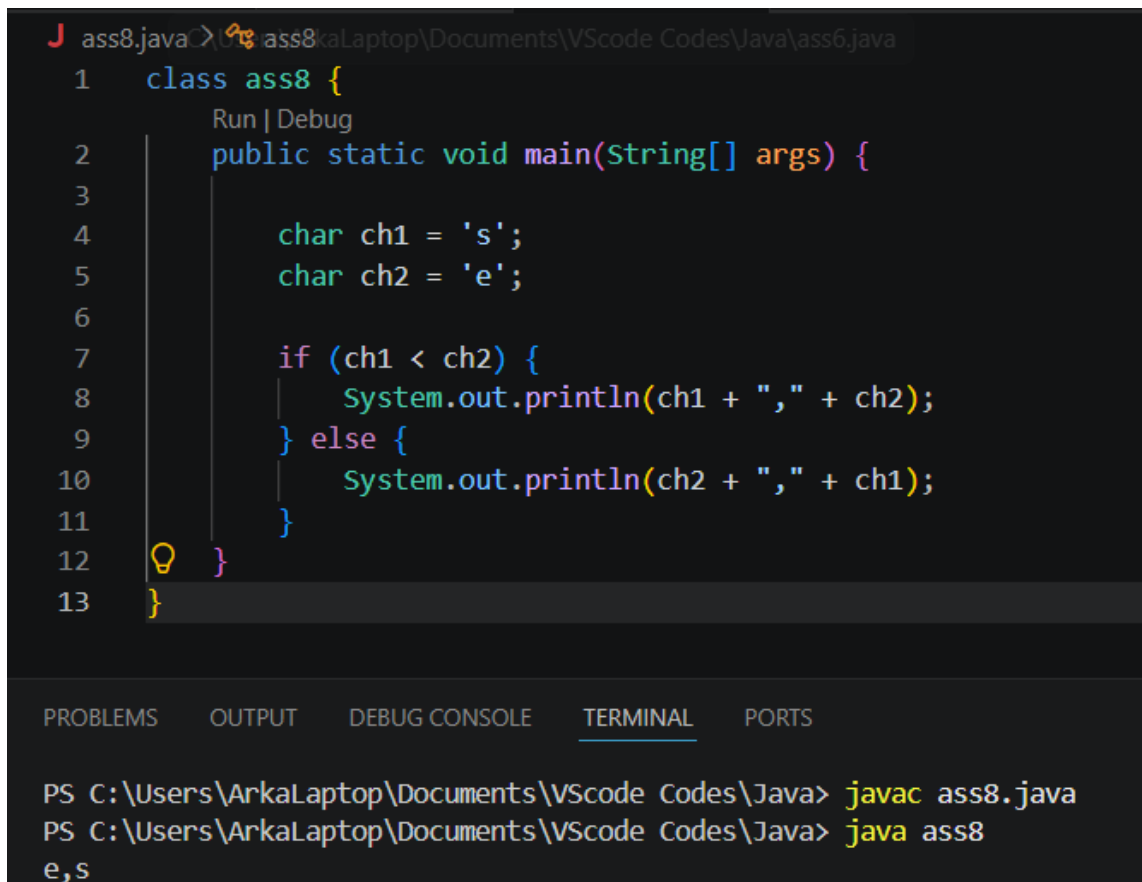
Q6. Write a Java program to check if the program has received **command-line arguments** or not. If the program has **not received any arguments**, print: No Values. Otherwise, print **all the arguments in a single line separated by commas (,)**.

```
ass7.java > ass7
1  class ass7 {
    Run | Debug
2      public static void main(String[] args) {
3
4          if (args.length == 0) {
5              System.out.println("No Values");
6          } else {
7              for (int i = 0; i < args.length; i++) {
8                  System.out.print(args[i]);
9
10                 if (i != args.length - 1) {
11                     System.out.print(",");
12                 }
13             }
14         }
15     }
16 }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\ArkaLaptop\Documents\VScode Codes\Java> javac ass7.java
PS C:\Users\ArkaLaptop\Documents\VScode Codes\Java> java ass7
No Values
PS C:\Users\ArkaLaptop\Documents\VScode Codes\Java> java ass7 Kolkata Delhi
Kolkata,Delhi
```

Q7. Initialize **two-character variables** in a Java program and display the characters in **alphabetical order**.



The screenshot shows a VS Code editor with a Java file named `ass8.java`. The code defines a class `ass8` with a `main` method. Inside the `main` method, two character variables `ch1` and `ch2` are initialized with the values 's' and 'e' respectively. An `if` statement checks if `ch1` is less than `ch2`. If true, it prints `ch1` followed by a comma and `ch2`. Otherwise, it prints `ch2` followed by a comma and `ch1`. The terminal at the bottom shows the compilation and execution of the program, resulting in the output `e,s`.

```
J ass8.java C:\ArkaLaptop\Documents\VScode Codes\Java\ass6.java
1  class ass8 {
    Run | Debug
2      public static void main(String[] args) {
3
4          char ch1 = 's';
5          char ch2 = 'e';
6
7          if (ch1 < ch2) {
8              System.out.println(ch1 + "," + ch2);
9          } else {
10             System.out.println(ch2 + "," + ch1);
11         }
12     }
13 }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\ArkaLaptop\Documents\VScode Codes\Java> javac ass8.java
PS C:\Users\ArkaLaptop\Documents\VScode Codes\Java> java ass8
e,s
```


Q8. Initialize a **character variable** in a Java program and:

- Print **"Alphabet"** if the initialized value is an alphabet.
- Print **"Digit"** if the initialized value is a number.
- Print **"Special Character"** if the initialized value is anything else.

```
J ass9.java > ass9
1 class ass9 {
    Run | Debug
2     public static void main(String[] args) {
3
4         char ch = 'A'; // Initialize the character
5
6         if ((ch >= 'A' && ch <= 'Z') || (ch >= 'a' && ch <= 'z')) {
7             System.out.println(x: "Alphabet");
8         } else if (ch >= '0' && ch <= '9') {
9             System.out.println(x: "Digit");
10        } else {
11            System.out.println(x: "Special Character");
12        }
13    }
14 }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\ArkaLaptop\Documents\VScode Codes\Java> javac ass9.java
PS C:\Users\ArkaLaptop\Documents\VScode Codes\Java> java ass9
Alphabet
```

```
J ass9.java > ass9
1 class ass9 {
    Run | Debug
2     public static void main(String[] args) {
3
4         char ch = '5'; // Initialize the character
5
6         if ((ch >= 'A' && ch <= 'Z') || (ch >= 'a' && ch <= 'z')) {
7             System.out.println(x: "Alphabet");
8         } else if (ch >= '0' && ch <= '9') {
9             System.out.println(x: "Digit");
10        } else {
11            System.out.println(x: "Special Character");
12        }
13    }
14 }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\ArkaLaptop\Documents\VScode Codes\Java> javac ass9.java
PS C:\Users\ArkaLaptop\Documents\VScode Codes\Java> java ass9
Digit
```

Q9. Write a Java program to accept **gender** ("Male" or "Female") and **age** from **command-line arguments** and print the **percentage of interest** based on the following conditions:

- If the gender is **Female** and age is between **1 and 58**, the interest is **8.2%**.
- If the gender is **Female** and age is between **59 and 100**, the interest is **9.2%**.
- If the gender is **Male** and age is between **1 and 58**, the interest is **8.4%**.
- If the gender is **Male** and age is between **59 and 100**, the interest is **10.5%**.

```
J ass10.java > ass10
1  class ass10 {
2      public static void main(String[] args) {
3
4          System.out.println(x: "Please pass Gender and Age as command line arguments.");
5          return;
6      }
7
8      String gender = args[0];
9      int age = Integer.parseInt(args[1]);
10
11      if (gender.equalsIgnoreCase(anotherString: "Female")) {
12          if (age >= 1 && age <= 58) {
13              System.out.println(x: "Percentage of Interest = 8.2%");
14          } else if (age >= 59 && age <= 100) {
15              System.out.println(x: "Percentage of Interest = 9.2%");
16          }
17      } else if (gender.equalsIgnoreCase(anotherString: "Male")) {
18          if (age >= 1 && age <= 58) {
19              System.out.println(x: "Percentage of Interest = 8.4%");
20          } else if (age >= 59 && age <= 100) {
21              System.out.println(x: "Percentage of Interest = 10.5%");
22          }
23      } else {
24          System.out.println(x: "Invalid Gender");
25      }
26  }
27  }
28  }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\ArkaLaptop\Documents\VScode Codes\Java> javac ass10.java
PS C:\Users\ArkaLaptop\Documents\VScode Codes\Java> java ass10 Female 20
Percentage of Interest = 8.2%
```

Q10. Initialize a **character variable** with an alphabet in a Java program.

- If the character is in **lowercase**, convert it to **uppercase** and display it in the format:

a->A

- If the character is in **uppercase**, convert it to **lowercase** and display it in the format:

A->a

```
J ass11.java > ass11
1  class ass11 {
    Run | Debug
2      public static void main(String[] args) {
3
4          char ch = 'a';    // Initialize the character
5
6          if (ch >= 'a' && ch <= 'z') {
7              System.out.println(ch + "->" + Character.toUpperCase(ch));
8          } else if (ch >= 'A' && ch <= 'Z') {
9              System.out.println(ch + "->" + Character.toLowerCase(ch));
10         } else {
11             System.out.println(x: "Invalid Input");
12         }
13     }
14 }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

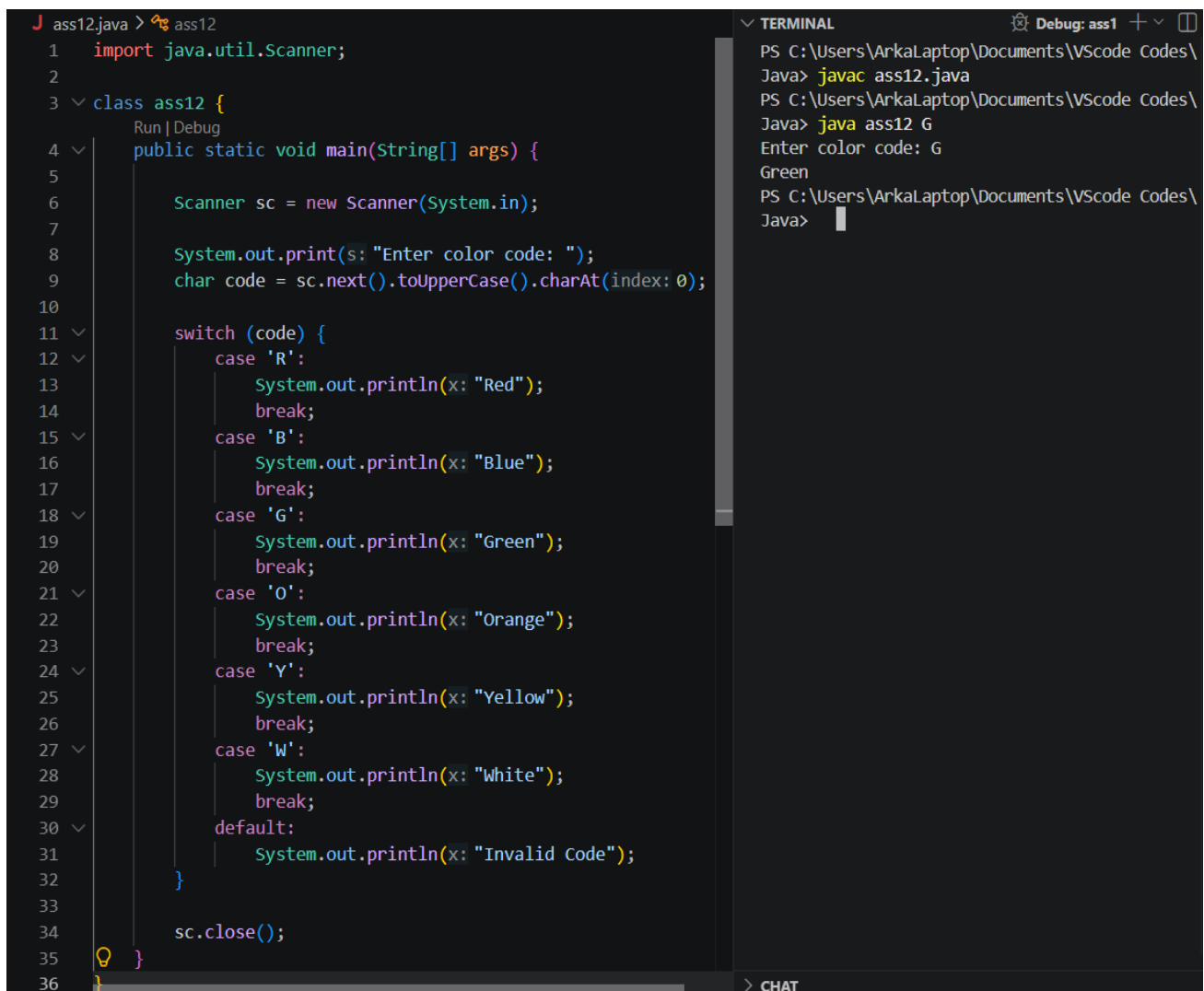
```
PS C:\Users\ArkaLaptop\Documents\VScode Codes\Java> javac ass11.java
PS C:\Users\ArkaLaptop\Documents\VScode Codes\Java> java ass11
a->A
```

Q11. Write a Java program to receive a **color code** from the user (an alphabet).

The program should print the corresponding **color name** based on the following codes:

- **R** → Red
- **B** → Blue
- **G** → Green
- **O** → Orange
- **Y** → Yellow
- **W** → White

If the color code entered by the user is **not valid**, print: Invalid Code



```
J ass12.java > ass12
1  import java.util.Scanner;
2
3  class ass12 {
    Run | Debug
4  public static void main(String[] args) {
5
6      Scanner sc = new Scanner(System.in);
7
8      System.out.print(s: "Enter color code: ");
9      char code = sc.next().toUpperCase().charAt(index: 0);
10
11     switch (code) {
12     case 'R':
13         System.out.println(x: "Red");
14         break;
15     case 'B':
16         System.out.println(x: "Blue");
17         break;
18     case 'G':
19         System.out.println(x: "Green");
20         break;
21     case 'O':
22         System.out.println(x: "Orange");
23         break;
24     case 'Y':
25         System.out.println(x: "Yellow");
26         break;
27     case 'W':
28         System.out.println(x: "White");
29         break;
30     default:
31         System.out.println(x: "Invalid Code");
32     }
33
34     sc.close();
35 }
36 }
```

```
TERMINAL
Debug: ass1
PS C:\Users\ArkaLaptop\Documents\VScode Codes\
Java> javac ass12.java
PS C:\Users\ArkaLaptop\Documents\VScode Codes\
Java> java ass12 G
Enter color code: G
Green
PS C:\Users\ArkaLaptop\Documents\VScode Codes\
Java>
```

> CHAT

Q12. Write a Java program to receive a **month number** as a **command-line argument** and print the corresponding **month name**.

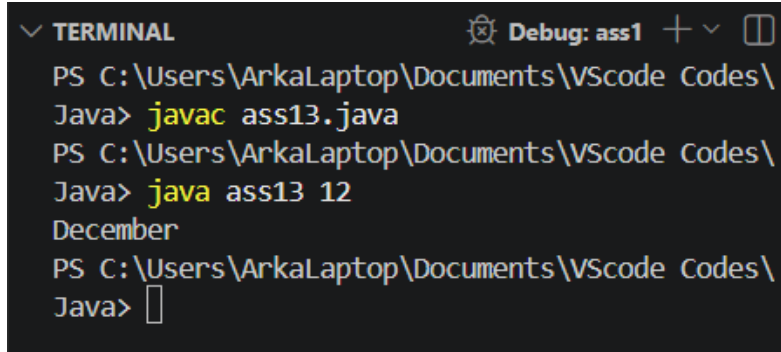
- If no command-line argument is provided, print: Please enter the month in numbers
- If the month number is not between **1 and 12**, print: Invalid month

Code:

```
class ass13 {  
  
    public static void main(String[] args) {  
  
        if (args.length == 0) {  
            System.out.println("Please enter the month in numbers");  
            return;  
        }  
  
        int month = Integer.parseInt(args[0]);  
  
        switch (month) {  
            case 1:  
                System.out.println("January");  
                break;  
            case 2:  
                System.out.println("February");  
                break;  
            case 3:  
                System.out.println("March");  
                break;  
            case 4:  
                System.out.println("April");  
                break;  
        }  
    }  
}
```

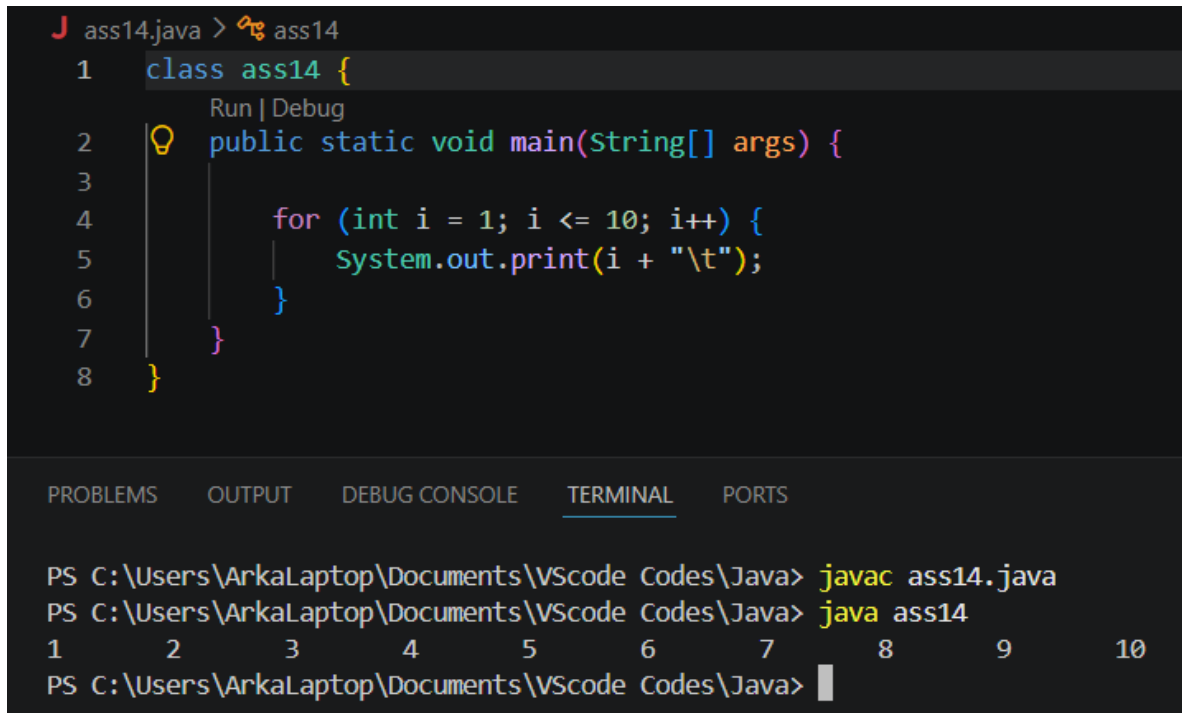
```
        case 5:
            System.out.println("May");
            break;
        case 6:
            System.out.println("June");
            break;
        case 7:
            System.out.println("July");
            break;
        case 8:
            System.out.println("August");
            break;
        case 9:
            System.out.println("September");
            break;
        case 10:
            System.out.println("October");
            break;
        case 11:
            System.out.println("November");
            break;
        case 12:
            System.out.println("December");
            break;
        default:
            System.out.println("Invalid month");
    }
}
```

Output:



```
✓ TERMINAL Debug: ass1 + ▢  
PS C:\Users\ArkaLaptop\Documents\VScode Codes\Java> javac ass13.java  
PS C:\Users\ArkaLaptop\Documents\VScode Codes\Java> java ass13 12  
December  
PS C:\Users\ArkaLaptop\Documents\VScode Codes\Java> 
```

Q13. Write a Java program to print numbers from 1 to 10 in a single row with one tab space between each number.



```
J ass14.java > ass14
1 class ass14 {
    Run | Debug
2 public static void main(String[] args) {
3
4     for (int i = 1; i <= 10; i++) {
5         System.out.print(i + "\t");
6     }
7 }
8 }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\ArkaLaptop\Documents\VScode Codes\Java> javac ass14.java
PS C:\Users\ArkaLaptop\Documents\VScode Codes\Java> java ass14
1      2      3      4      5      6      7      8      9      10
PS C:\Users\ArkaLaptop\Documents\VScode Codes\Java>
```


Q14. Write a Java program to print **even numbers between 23 and 57**. Each number should be printed on a **separate line**.

```
J ass15.java >  ass15
1  class ass15 {
    Run | Debug
2      public static void main(String[] args) {
3
4          for (int i = 23; i <= 57; i++) {
5              if (i % 2 == 0) {
6                  System.out.println(i);
7              }
8          }
9      }
10 }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\ArkaLaptop\Documents\VScode Codes\Java> javac ass15.java
PS C:\Users\ArkaLaptop\Documents\VScode Codes\Java> java ass15
24
26
28
30
32
34
36
38
40
42
44
46
48
50
52
54
56
```

Q15. Write a Java program to check whether a given number is Prime or Not Prime.

```
J ass16.java > ass16 > main(String[])
1  import java.util.Scanner;
2
3  class ass16 {
4      public static void main(String[] args) {
5
6          Scanner sc = new Scanner(System.in);
7
8          System.out.print(s: "Enter a number: ");
9          int num = sc.nextInt();
10
11         boolean isPrime = true;
12
13         if (num <= 1) {
14             isPrime = false;
15         } else {
16             for (int i = 2; i <= num / 2; i++) {
17                 if (num % i == 0) {
18                     isPrime = false;
19                     break;
20                 }
21             }
22         }
23
24         if (isPrime) {
25             System.out.println(num + " is a Prime Number");
26         } else {
27             System.out.println(num + " is Not a Prime Number");
28         }
29
30         sc.close();
31     }
32 }
```

TERMINAL

```
PS C:\Users\ArkaLaptop\Documents\VScode Codes\Java> javac ass16.java
PS C:\Users\ArkaLaptop\Documents\VScode Codes\Java> java ass16
Enter a number: 156
156 is Not a Prime Number
PS C:\Users\ArkaLaptop\Documents\VScode Codes\Java>
```

Q16. Write a Java program to print **all prime numbers between 10 and 99**.

```
J ass17.java >  ass17
1  class ass17 {
2      Run | Debug
3      public static void main(String[] args) {
4          for (int num = 10; num <= 99; num++) {
5              boolean isPrime = true;
6
7              for (int i = 2; i <= num / 2; i++) {
8                  if (num % i == 0) {
9                      isPrime = false;
10                     break;
11                 }
12             }
13
14             if (isPrime) {
15                 System.out.println(num);
16             }
17         }
18     }
19 }
```

PROBLEMS OUTPUT TERMINAL DEBUG CONSOLE PORTS

```
PS C:\Users\ArkaLaptop\Documents\VScode Codes\Java> javac ass17.java
PS C:\Users\ArkaLaptop\Documents\VScode Codes\Java> java ass17
11
13
17
19
23
29
```

Q17. Write a Java program to print the **sum of all the digits** of a given number.

```
J ass18.java >  ass18
1  import java.util.Scanner;
2
3  class ass18 {
    Run | Debug
4      public static void main(String[] args) {
5
6          Scanner sc = new Scanner(System.in);
7
8          System.out.print(s: "Enter a number: ");
9          int num = sc.nextInt();
10
11         int sum = 0;
12
13         while (num > 0) {
14             int digit = num % 10;
15             sum = sum + digit;
16             num = num / 10;
17         }
18
19         System.out.println("Sum of digits = " + sum);
20
21         sc.close();
22     }
23 }
```

PROBLEMS OUTPUT TERMINAL DEBUG CONSOLE PORTS

```
PS C:\Users\ArkaLaptop\Documents\VScode Codes\Java> javac ass18.java
PS C:\Users\ArkaLaptop\Documents\VScode Codes\Java> java ass18
Enter a number: 56
Sum of digits = 11
```

Q18. Write a Java program to print * in **Floyd's format** (using for and while loops).

```
J ass19.java >  ass19
1  class ass19 {
    Run | Debug
2  public static void main(String[] args) {
3
4      if (args.length == 0) {
5          System.out.println(x: "Please enter an integer number");
6          return;
7      }
8
9      int n = Integer.parseInt(args[0]);
10
11     for (int i = 1; i <= n; i++) {
12         for (int j = 1; j <= i; j++) {
13             System.out.print(s: "* ");
14         }
15         System.out.println();
16     }
17 }
18 }
```

PROBLEMS OUTPUT TERMINAL DEBUG CONSOLE PORTS

```
PS C:\Users\ArkaLaptop\Documents\VScode Codes\Java> javac ass19.java
PS C:\Users\ArkaLaptop\Documents\VScode Codes\Java> java ass19 6
*
* *
* * *
* * * *
* * * * *
* * * * * *
```

Q19. Write a Java program to **reverse a given number** and print the reversed number.

```
J ass20.java >  ass20
1  import java.util.Scanner;
2  
3  class ass20 {
4      Run | Debug
      public static void main(String[] args) {
5
6          Scanner sc = new Scanner(System.in);
7
8          System.out.print(s: "Enter a number: ");
9          int num = sc.nextInt();
10
11         int reverse = 0;
12
13         while (num > 0) {
14             int digit = num % 10;
15             reverse = reverse * 10 + digit;
16             num = num / 10;
17         }
18
19         System.out.println("Reversed Number = " + reverse);
20
21         sc.close();
22     }
23 }
```

PROBLEMS OUTPUT TERMINAL DEBUG CONSOLE PORTS

```
PS C:\Users\ArkaLaptop\Documents\VScode Codes\Java> javac ass20.java
PS C:\Users\ArkaLaptop\Documents\VScode Codes\Java> java ass20
Enter a number: 20
Reversed Number = 2
PS C:\Users\ArkaLaptop\Documents\VScode Codes\Java> 
```

Q20. Write a Java program to find whether a given number is a **Palindrome** or not.

```
J ass21.java >  ass21
1  class ass21 {
    Run | Debug
2      public static void main(String[] args) {
3
4          if (args.length == 0) {
5              System.out.println(x: "Please enter a number");
6              return;
7          }
8
9          int num = Integer.parseInt(args[0]);
10         int original = num;
11         int reverse = 0;
12
13         while (num > 0) {
14             int digit = num % 10;
15             reverse = reverse * 10 + digit;
16             num = num / 10;
17         }
18
19         if (original == reverse) {
20             System.out.println(original + " is a palindrome");
21         } else {
22             System.out.println(original + " is not a palindrome");
23         }
24     }
25 }
```

PROBLEMS OUTPUT TERMINAL DEBUG CONSOLE PORTS

```
PS C:\Users\ArkaLaptop\Documents\VScode Codes\Java> javac ass21.java
PS C:\Users\ArkaLaptop\Documents\VScode Codes\Java> java ass21 110011
110011 is a palindrome
```

Arrays

Q21. Write a Java program to **initialize an integer array** and print the **sum** and **average** of the array elements.

```
ass22.java > ass22
1  class ass22 {
    Run | Debug
2  public static void main(String[] args) {
3
4      // Initialize the array
5      int[] arr = {10, 20, 30, 40, 50};
6
7      int sum = 0;
8
9      // Calculate sum
10     for (int i = 0; i < arr.length; i++) {
11         sum = sum + arr[i];
12     }
13
14     // Calculate average
15     double average = (double) sum / arr.length;
16
17     System.out.println("Sum = " + sum);
18     System.out.println("Average = " + average);
19 }
20 }
```

PROBLEMS OUTPUT TERMINAL DEBUG CONSOLE PORTS

```
PS C:\Users\ArkaLaptop\Documents\VScode Codes\Java> javac ass22.java
PS C:\Users\ArkaLaptop\Documents\VScode Codes\Java> java ass22
Sum = 150
Average = 30.0
```


Q22. Write a Java program to **initialize an integer array** and find the **maximum** and **minimum** value of the array.

```
J ass23.java >  ass23
1  class ass23 {
2      Run | Debug
3       public static void main(String[] args) {
4          // Initialize the array
5          int[] arr = {25, 10, 45, 5, 30};
6
7          // Assume first element is both maximum and minimum
8          int max = arr[0];
9          int min = arr[0];
10
11         // Find maximum and minimum
12         for (int i = 1; i < arr.length; i++) {
13
14             if (arr[i] > max) {
15                 max = arr[i];
16             }
17
18             if (arr[i] < min) {
19                 min = arr[i];
20             }
21         }
22
23         System.out.println("Maximum = " + max);
24         System.out.println("Minimum = " + min);
25     }
26 }
```

PROBLEMS OUTPUT TERMINAL DEBUG CONSOLE PORTS

```
PS C:\Users\ArkaLaptop\Documents\VScode Codes\Java> javac ass23.java
PS C:\Users\ArkaLaptop\Documents\VScode Codes\Java> java ass23
Maximum = 45
Minimum = 5
```

Q23. Write a Java program to **initialize an integer array** and check if a given number is present in the array.

- If the number is **not found**, print **-1**.
- Otherwise, print the **index** of the given number in the array.

```
J ass24.java >  ass24
1  class ass24 {
2      Run | Debug
3       public static void main(String[] args) {
4          // Initialize the array
5          int[] arr = {1, 4, 34, 56, 7};
6
7          // Search element
8          int search = 56;
9
10         int index = -1;
11
12         // Search for the element
13         for (int i = 0; i < arr.length; i++) {
14             if (arr[i] == search) {
15                 index = i;
16                 break;
17             }
18         }
19
20         System.out.println(index);
21     }
22 }
```

PROBLEMS OUTPUT TERMINAL DEBUG CONSOLE PORTS

```
PS C:\Users\ArkaLaptop\Documents\VScode Codes\Java> javac ass24.java
PS C:\Users\ArkaLaptop\Documents\VScode Codes\Java> java ass24
3
```

Q24. Write a Java program to **initialize an integer array with ASCII values** and print the **corresponding character values** in a single row.

```
J ass25.java > ass25
1  class ass25 {
    Run | Debug
2  public static void main(String[] args) {
3
4      // Initialize the array with ASCII values
5      int[] arr = {65, 66, 67, 68, 69};
6
7      // Print corresponding characters
8      for (int i = 0; i < arr.length; i++) {
9          System.out.print((char) arr[i] + " ");
10     }
11 }
12 }
```

PROBLEMS OUTPUT TERMINAL DEBUG CONSOLE PORTS

```
PS C:\Users\ArkaLaptop\Documents\VScode Codes\Java> javac ass25.java
PS C:\Users\ArkaLaptop\Documents\VScode Codes\Java> java ass25
A B C D E
PS C:\Users\ArkaLaptop\Documents\VScode Codes\Java> 
```

Q25. Write a Java program to find the **largest 2 numbers** and the **smallest 2 numbers** in a given integer array.

```
J ass26.java >  ass26
1  import java.util.Arrays;
2  
3  class ass26 {
    Run | Debug
4      public static void main(String[] args) {
5
6          // Initialize the array
7          int[] arr = {5, 2, 8, 1, 9, 3};
8
9          // Sort the array
10         Arrays.sort(arr);
11
12         System.out.println(x: "Smallest 2 numbers:");
13         System.out.println(arr[0]);
14         System.out.println(arr[1]);
15
16         System.out.println(x: "Largest 2 numbers:");
17         System.out.println(arr[arr.length - 2]);
18         System.out.println(arr[arr.length - 1]);
19     }
20 }
```

PROBLEMS OUTPUT TERMINAL DEBUG CONSOLE PORTS

```
PS C:\Users\ArkaLaptop\Documents\VScode Codes\Java> javac ass26.java
PS C:\Users\ArkaLaptop\Documents\VScode Codes\Java> java ass26
Smallest 2 numbers:
1
2
Largest 2 numbers:
8
9
```

Q26. Write a Java program to **initialize an integer array** and print the elements in **sorted order**.

```
J ass27.java > 🐞 ass27 > 📄 main(String[])
1  import java.util.Arrays;
2
3  class ass27 {
4      public static void main(String[] args) {
5
6          // Initialize the array
7          int[] arr = {25, 10, 45, 5, 30};
8
9          // Sort the array
10         Arrays.sort(arr);
11
12         // Print the sorted array
13         System.out.println("Sorted Array:");
14
15         for (int i = 0; i < arr.length; i++) {
16             System.out.print(arr[i] + " ");
17         }
18     }
19 }
```

PROBLEMS OUTPUT TERMINAL DEBUG CONSOLE PORTS

```
PS C:\Users\ArkaLaptop\Documents\VScode Codes\Java> javac ass27.java
PS C:\Users\ArkaLaptop\Documents\VScode Codes\Java> java ass27
Sorted Array:
5 10 25 30 45
```

Q27. Write a Java program to **remove the duplicate elements** from an integer array and print the resulting array.

```
ass28.java > ass28
1  class ass28 {
    Run | Debug
2      public static void main(String[] args) {
3
4          // Initialize the array
5          int[] arr = {12, 34, 12, 45, 67, 89};
6
7          System.out.print(s: "Array after removing duplicates: ");
8
9          for (int i = 0; i < arr.length; i++) {
10             boolean isDuplicate = false;
11
12             // Check if arr[i] has appeared before
13             for (int j = 0; j < i; j++) {
14                 if (arr[i] == arr[j]) {
15                     isDuplicate = true;
16                     break;
17                 }
18             }
19
20             if (!isDuplicate) {
21                 System.out.print(arr[i] + " ");
22             }
23         }
24     }
25 }
```

PROBLEMS OUTPUT TERMINAL DEBUG CONSOLE PORTS powershell ⚠

```
PS C:\Users\ArkaLaptop\Documents\VScode Codes\Java> javac ass28.java
PS C:\Users\ArkaLaptop\Documents\VScode Codes\Java> java ass28
Array after removing duplicates: 12 34 45 67 89
```

Q28. Write a Java program to print the **sum of the elements of an array** following the given condition:

- If the array contains **6** followed by **7**, ignore **6**, **7**, and **all the numbers between them** while calculating the sum.
- Add only the remaining elements.

```
J ass29.java > ass29
1 class ass29 {
2     public static void main(String[] args) {
3
4         // Initialize the array
5         int[] arr = {10, 3, 6, 1, 2, 7, 9};
6
7         int sum = 0;
8         boolean ignore = false;
9
10        for (int i = 0; i < arr.length; i++) {
11
12            if (arr[i] == 6) {
13                ignore = true;
14            }
15
16            if (!ignore) {
17                sum = sum + arr[i];
18            }
19
20            if (ignore && arr[i] == 7) {
21                ignore = false;
22            }
23        }
24
25        System.out.println("Sum = " + sum);
26    }
27 }
```

PROBLEMS OUTPUT TERMINAL DEBUG CONSOLE PORTS

```
PS C:\Users\ArkaLaptop\Documents\VScode Codes\Java> javac ass29.java
PS C:\Users\ArkaLaptop\Documents\VScode Codes\Java> java ass29
Sum = 22
```

Q29. Write a Java program to print a version of the given array where **all the 10's have been removed**.

- The remaining elements should be shifted to the **left**.
- The empty spaces at the end of the array should be filled with **0**.
- You may modify the original array or create a new array.

```
J ass30.java > ass30
1 class ass30 {
    Run | Debug
2     public static void main(String[] args) {
3
4         // Initialize the array
5         int[] arr = {1, 10, 10, 2};
6
7         int[] result = new int[arr.length];
8         int index = 0;
9
10        // Copy all elements except 10
11        for (int i = 0; i < arr.length; i++) {
12            if (arr[i] != 10) {
13                result[index] = arr[i];
14                index++;
15            }
16        }
17
18        // Print the new array
19        System.out.print(s: "Output: ");
20
21        for (int i = 0; i < result.length; i++) {
22            System.out.print(result[i] + " ");
23        }
24    }
25 }
```

PROBLEMS OUTPUT TERMINAL DEBUG CONSOLE PORTS

```
PS C:\Users\ArkaLaptop\Documents\VScode Codes\Java> javac ass30.java
PS C:\Users\ArkaLaptop\Documents\VScode Codes\Java> java ass30
Output: 1 2 0 0
```


Q30. Write a Java program to print an array that contains the **same numbers** as the given array, but rearranged so that **all the even numbers come before all the odd numbers**.

- The order among even numbers or among odd numbers does **not matter**.
- You may modify the original array or create a new array.

```
J ass31.java > ...
1  class ass31 {
    Run | Debug
2      public static void main(String[] args) {
3
4          // Initialize the array
5          int[] arr = {1, 0, 1, 0, 0, 1, 1};
6
7          int[] result = new int[arr.length];
8          int index = 0;
9
10         // Copy even numbers first
11         for (int i = 0; i < arr.length; i++) {
12             if (arr[i] % 2 == 0) {
13                 result[index] = arr[i];
14                 index++;
15             }
16         }
17
18         // Copy odd numbers
19         for (int i = 0; i < arr.length; i++) {
20             if (arr[i] % 2 != 0) {
21                 result[index] = arr[i];
22                 index++;
23             }
24         }
25
26         // Print the result array
27         System.out.print(s: "Output: ");
28
29         for (int i = 0; i < result.length; i++) {
30             System.out.print(result[i] + " ");
31         }
32     }
33 }
```

TERMINAL powershell

```
PS C:\Users\ArkaLaptop\Documents\VScode C
odes\Java> javac ass31.java
PS C:\Users\ArkaLaptop\Documents\VScode C
odes\Java> java ass31
Output: 0 0 0 1 1 1 1
PS C:\Users\ArkaLaptop\Documents\VScode C
odes\Java>
```

CHAT SESSIONS

Q31. Given an integer array, print **true** if **every element** in the array is either **1** or **4**. Otherwise, print **false**.

```
J ass32.java > ass32 > main(String[])
1  class ass32 {
    Run | Debug
2      public static void main(String[] args) {
3
4          // Initialize the array
5          int[] arr = {1, 4, 1, 4};
6
7          boolean flag = true;
8
9          for (int i = 0; i < arr.length; i++) {
10             if (arr[i] != 1 && arr[i] != 4) {
11                 flag = false;
12                 break;
13             }
14         }
15
16         System.out.println(flag);
17     }
18 }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\ArkaLaptop\Documents\VScode Codes\Java> javac ass32.java
PS C:\Users\ArkaLaptop\Documents\VScode Codes\Java> java ass32
true
```

Q32. Given two integer arrays, a and b, each of length 3, create a new array of length 2 containing the **middle element** of each array.

```
J ass33.java >  ass33
1  class ass33 {
    Run | Debug
2      public static void main(String[] args) {
3
4          // Initialize the arrays
5          int[] a = {1, 2, 3};
6          int[] b = {4, 5, 6};
7
8          // Create a new array containing the middle elements
9          int[] result = {a[1], b[1]};
10
11         // Print the result
12         System.out.print(s: "Output: ");
13
14         for (int i = 0; i < result.length; i++) {
15             System.out.print(result[i] + " ");
16         }
17     }
18 }
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\ArkaLaptop\Documents\VScode Codes\Java> javac ass33.java
PS C:\Users\ArkaLaptop\Documents\VScode Codes\Java> java ass33
Output: 2 5
```

Q33. Write a Java program to **reverse the elements of a given 2×2 array**.

- Four integer numbers must be passed as **command-line arguments**.
- If fewer than four arguments are passed, print: Please enter 4 integer numbers.

```
J ass34.java > ass34
1 class ass34 {
  Run | Debug
2   public static void main(String[] args) {
3
4       if (args.length != 4) {
5           System.out.println(x: "Please enter 4 integer numbers");
6           return;
7       }
8
9       int[][] arr = new int[2][2];
10      int k = 0;
11      // Fill the 2x2 array
12      for (int i = 0; i < 2; i++) {
13          for (int j = 0; j < 2; j++) {
14              arr[i][j] = Integer.parseInt(args[k]);
15              k++;
16          }
17      }
18      // Print original array
19      System.out.println(x: "The given array is :");
20      for (int i = 0; i < 2; i++) {
21          for (int j = 0; j < 2; j++) {
22              System.out.print(arr[i][j] + " ");
23          }
24          System.out.println();
25      }
26      // Print reversed array
27      System.out.println(x: "\nThe reverse of the array is :");
28      for (int i = 1; i >= 0; i--) {
29          for (int j = 1; j >= 0; j--) {
30              System.out.print(arr[i][j] + " ");
31          }
32          System.out.println();
33      }
34  }
35 }
```

TERMINAL

```
PS C:\Users\ArkaLaptop\Documents\VScode C
odes\Java> javac ass34.java
PS C:\Users\ArkaLaptop\Documents\VScode C
odes\Java> java ass34 1 2 3 4
The given array is :
1 2
3 4

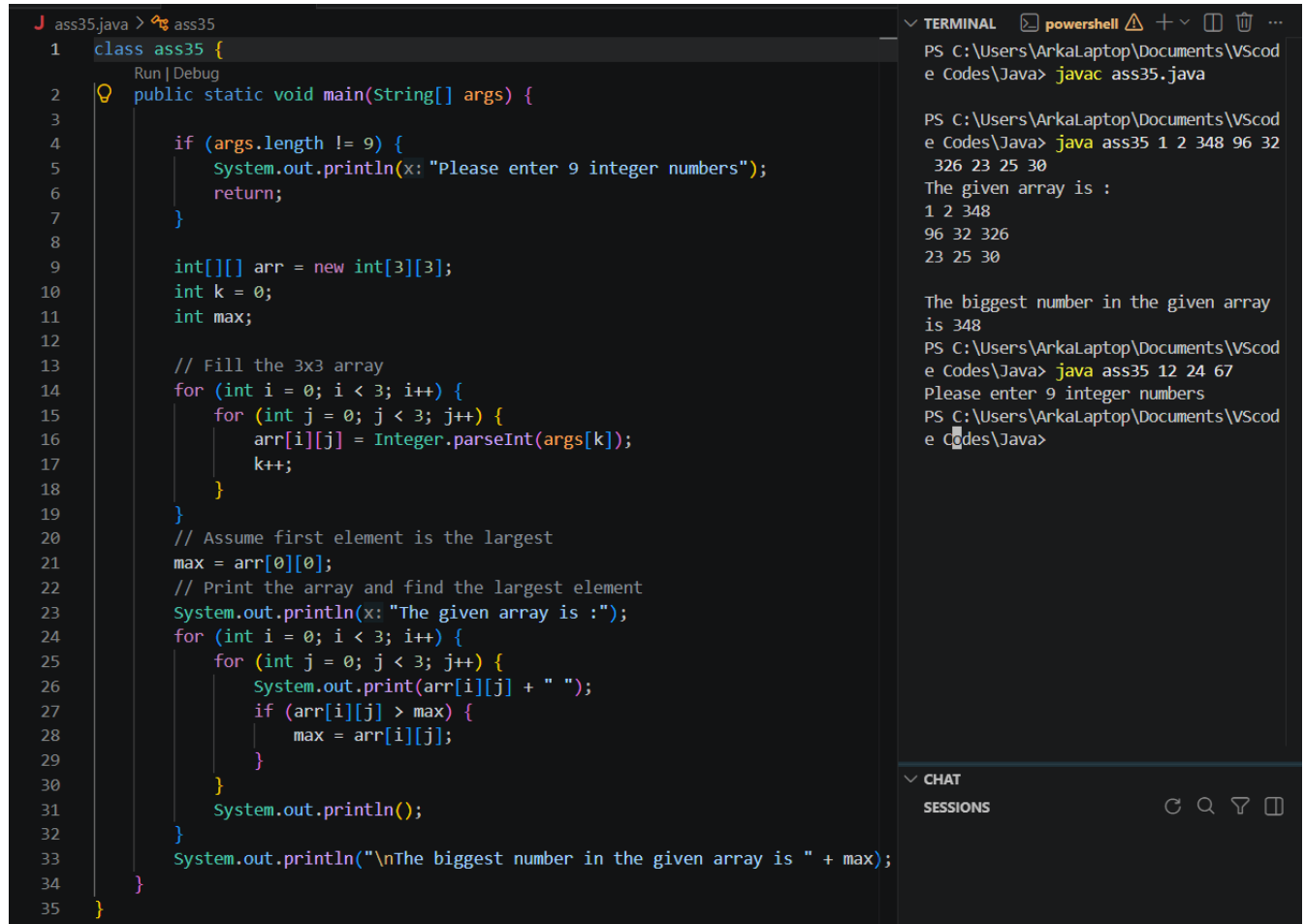
The reverse of the array is :
4 3
2 1
PS C:\Users\ArkaLaptop\Documents\VScode C
odes\Java> java ass34 1 2 3
Please enter 4 integer numbers
PS C:\Users\ArkaLaptop\Documents\VScode C
odes\Java>
```

CHAT

SESSIONS

Q34. Write a Java program to find the **biggest number** in a **3×3 array**.

- The program should receive **9 integer numbers** as **command-line arguments**.
- If fewer than **9 arguments** are passed, print: Please enter 9 integer numbers



```
1 class ass35 {
2     public static void main(String[] args) {
3
4         if (args.length != 9) {
5             System.out.println(x: "Please enter 9 integer numbers");
6             return;
7         }
8
9         int[][] arr = new int[3][3];
10        int k = 0;
11        int max;
12
13        // Fill the 3x3 array
14        for (int i = 0; i < 3; i++) {
15            for (int j = 0; j < 3; j++) {
16                arr[i][j] = Integer.parseInt(args[k]);
17                k++;
18            }
19        }
20        // Assume first element is the largest
21        max = arr[0][0];
22        // Print the array and find the largest element
23        System.out.println(x: "The given array is :");
24        for (int i = 0; i < 3; i++) {
25            for (int j = 0; j < 3; j++) {
26                System.out.print(arr[i][j] + " ");
27                if (arr[i][j] > max) {
28                    max = arr[i][j];
29                }
30            }
31            System.out.println();
32        }
33        System.out.println("\nThe biggest number in the given array is " + max);
34    }
35 }
```

TERMINAL

```
PS C:\Users\ArkaLaptop\Documents\VScode Codes\Java> javac ass35.java
PS C:\Users\ArkaLaptop\Documents\VScode Codes\Java> java ass35 1 2 348 96 32 326 23 25 30
The given array is :
1 2 348
96 32 326
23 25 30
The biggest number in the given array is 348
PS C:\Users\ArkaLaptop\Documents\VScode Codes\Java> java ass35 12 24 67
Please enter 9 integer numbers
PS C:\Users\ArkaLaptop\Documents\VScode Codes\Java>
```

CHAT

SESSIONS

Module Mini Project

Stream : JAVA

Tech Module : 1

Project:1

Given the following table containing information about employees of an organization, develop a small java application, which accepts employee id from the command prompt and displays the following details as output:

EmpNo EmpName Department Designation Salary

In your program, you must initialize an array with the following details.

Emp No	Emp Name	Join Date	Designation Code	Department	Basic	HRA	IT
1001	Ashish	01/04/2009	e	R&D	20000	8000	3000
1002	Sushma	23/08/2012	c	PM	30000	12000	9000
1003	Rahul	12/11/2008	k	Acct	10000	8000	1000
1004	Chahat	29/01/2013	r	Front Desk	12000	6000	2000
1005	Ranjan	16/07/2005	m	Engg	50000	20000	20000
1006	Suman	1/1/2000	e	Manufacturing	23000	9000	4400
1007	Tanmay	12/06/2006	c	PM	29000	12000	10000

DA (Dearness Allowance) details are given in the below "Designation" table.

Designation Code	Designation	DA
e	Engineer	20000
c	Consultant	32000
k	Clerk	12000
r	Receptionist	15000
m	Manager	40000

Note 1: Salary should be calculated as (Basic + HRA + DA – IT).

Note 2: Use switch-case to print Designation and to find the value of DA for a particular employee.

Expected Output format: (assuming that your class name is Project1)

- If you execute the command **java Project1 1003**, the output should be –

Emp No. Emp Name Department Designation Salary
1003 Rahul Acct Clerk 29000

- If you execute the command **java Project1 123**, the output should be –

There is no employee with empid : 123

Code:

```
class Project1 {

    public static void main(String args[]) {

        if (args.length != 1) {

            System.out.println("Please enter employee id");

            return;

        }

        int empId = Integer.parseInt(args[0]);

        int empNo[] = {1001,1002,1003,1004,1005,1006,1007};

        String empName[] = {

            "Ashish","Sushma","Rahul",

            "Chahat","Ranjan","Suman","Tanmay"

        };

        String department[] = {

            "R&D","PM","Acct",

            "Front Desk","Engg","Manufacturing","PM"

        };

        char designationCode[] = {

            'e','c','k','r','m','e','c'

        };

        int basic[] = {

            20000,30000,10000,
```

```
12000,50000,23000,29000  
};
```

```
int hra[] = {  
    8000,12000,8000,  
    6000,20000,9000,12000  
};
```

```
int it[] = {  
    3000,9000,1000,  
    2000,20000,4400,10000  
};
```

```
int index = -1;
```

```
for(int i=0;i<empNo.length;i++) {  
    if(empNo[i]==empId) {  
        index=i;  
        break;  
    }  
}
```

```
if(index==-1) {  
    System.out.println("There is no employee with empid : " + empId);  
    return;  
}
```

```
String designation="";
```

```
int da=0;
```



```
switch(designationCode[index]) {

    case 'e':

        designation="Engineer";

        da=20000;

        break;

    case 'c':

        designation="Consultant";

        da=32000;

        break;

    case 'k':

        designation="Clerk";

        da=12000;

        break;

    case 'r':

        designation="Receptionist";

        da=15000;

        break;

    case 'm':

        designation="Manager";

        da=40000;

        break;

}

int salary = basic[index] + hra[index] + da - it[index];
```

```
System.out.println("Emp No.\tEmp Name\tDepartment\tDesignation\tSalary");
```

```
System.out.println(  
    empNo[index] + "\t" +  
    empName[index] + "\t\t" +  
    department[index] + "\t\t" +  
    designation + "\t\t" +  
    salary);  
}  
}
```

Output:

```
✓ TERMINAL powershell + - [ ] [ ] ...  
PS C:\Users\ArkaLaptop\Documents\VScode Codes\Java> javac proj1.java  
PS C:\Users\ArkaLaptop\Documents\VScode Codes\Java> java proj1 1003  
Emp No. Emp Name      Department  Designation  Salary  
1003    Rahul              Acct       Clerk        29000  
PS C:\Users\ArkaLaptop\Documents\VScode Codes\Java> java proj1 1005  
Emp No. Emp Name      Department  Designation  Salary  
1005    Ranjan            Engg       Manager      90000  
PS C:\Users\ArkaLaptop\Documents\VScode Codes\Java> java proj1 123  
There is no employee with empid : 123  
PS C:\Users\ArkaLaptop\Documents\VScode Codes\Java> |
```